

Engaging Students with Instructional Strategies Mapped Against the Skill/Will Matrix

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Biography

Edwina Cheng is a Year Head and mathematics educator at Compassvale Secondary School in Singapore. Over her 19 years of teaching, she has held leadership roles such as Level Head for Student Management (2010–2011) and Subject Head for Student Leadership (2011–2018).

Edwina is passionate about supporting students with additional needs. She believes every child wants to feel respected and to learn, achieve, and be appreciated. She also believes in providing a learning environment with high support and high expectations. She believes in designing classroom experiences that make learning enjoyable. For her work, she received the Outstanding Youth in Education Award in 2011 and the President's Award for Teaching in 2021. Edwina is now in transition to become a teacher leader, where she aims to inspire and grow others in educational support.

Executive Summary

This educator project aims to apply the Will and Skill Matrix to identify the characteristics and behaviours of four types of students: high will/high skill, high will/low skill, low will/high skill, and low will/low skill. It also examines the effectiveness of various behavioural and cognitive instructional strategies in engaging each of these student types. Ultimately, the project seeks to help educators recognise students based on their will and skill levels and adopt the most suitable strategies to engage them. By doing so, our students stay engaged in their learning, develop motivation to learn, and achieve academic success.

Overview of Topic Area

Alex (pseudonym), a 15-year-old secondary school student, graduated from primary school with a Primary School Leaving Examinations Achievement Level of 14 out of 32. His parents enrolled him in a school near his home without considering his opinion. The school he went to accepted students with Achievement Levels between 14 to 18, making Alex one of the top scorers upon entry. One might expect Alex to be among the top students in his cohort. However, by the end of Secondary Three, Alex was identified as a low-progress learner, and his teachers found it challenging to manage him in class.

Alex often slept during lessons and rarely made any effort to participate in class activities. When he was awake, he tended to talk to his partner instead of paying attention. When his teacher intervened, he frequently commented that he already knew what the teacher was teaching, before returning to either disturb his friends or fall asleep again.

As assessments approached, Alex started to pay attention in class and often asked his teacher whether the content taught would be tested. Despite his disruptive behaviour and poor learning attitude, Alex managed to perform well in his assessments during Secondary One.

In Secondary Two, Alex continued to show a poor learning attitude and remained disruptive and disengaged during lessons. Nevertheless, he managed to pass his summative assessments and progress to Secondary Three. However, his learning attitude remained unchanged, and he struggled to keep up with the rigour of the upper secondary curriculum. He eventually failed to meet the promotion criteria. Alex represents a student who is low will/high skill.

Mrs. Toh (pseudonym), a secondary school teacher, taught students who were learning Mathematics at a less demanding level. As many of her students had a weak foundation in Mathematics, she designed engaging lessons to make learning enjoyable. One of her students, Lisa (pseudonym), enjoyed her lessons. Lisa paid close attention in class, frequently taking notes and consistently completing all her homework. She regularly revised her work after school using the resources provided by Mrs Toh, and frequently sought help from Mrs. Toh and her friends.

Mrs. Toh recognised Lisa as a hardworking student, even though she struggled and had been failing all her Mathematics assessments. Despite her repeated setbacks, she persevered and continued to put in her best effort. Mrs. Toh often felt guilty for not being able to help Lisa achieve success in Mathematics. Lisa represents a student who is high will/low skill.

Teachers often encounter students like Alex and Lisa in their classrooms, in addition to those who either excel academically or demonstrate low motivation to learn. Therefore, teachers work with students who possess different levels of will and skill, which affects engagement and learning outcomes, and may further impact the development of students' will and skill (Skinner et al., 2016). This project seeks to help educators recognise students based on their will and skill levels and adopt the most suitable strategies to engage them.

Background

Engagement and Motivation of Singapore Students

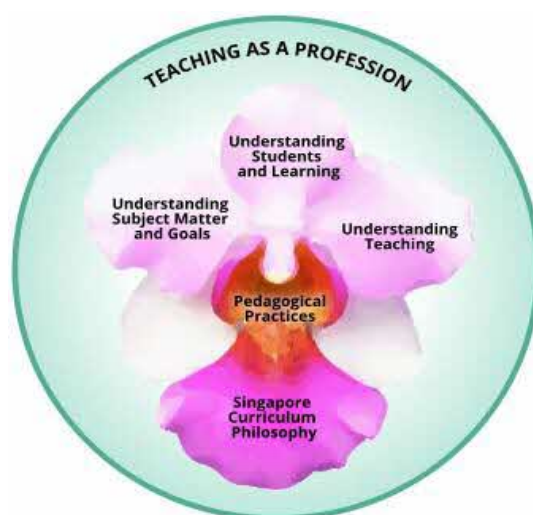
According to the Ministry of Education Singapore (MOES) (2025), the Singapore Curriculum Philosophy values every child as an individual, grounded in the belief that every

child wants to and can learn. As every child differs in their learning needs, experiences, beliefs, knowledge, and skills, effective learning requires teachers to adapt their teaching pace, approaches, and assessment practices to be developmentally appropriate and meaningful. When children find learning meaningful, they become engaged, motivated, challenged, and take ownership of their learning. These beliefs guide the Singapore teachers in the design and implementation of a curriculum that focuses on learner engagement.

Anchored in the Singapore Curriculum Philosophy, the Singapore Teaching Practice (STP) model was introduced by MOES (2025). As shown in Figure 1, the teaching and learning model is represented in the form of an orchid, the national flower of Singapore, symbolising the beliefs and pedagogical practices that are customised to guide and strengthen effective teaching for learner engagement.

Figure 1

The Singapore Teaching Practice

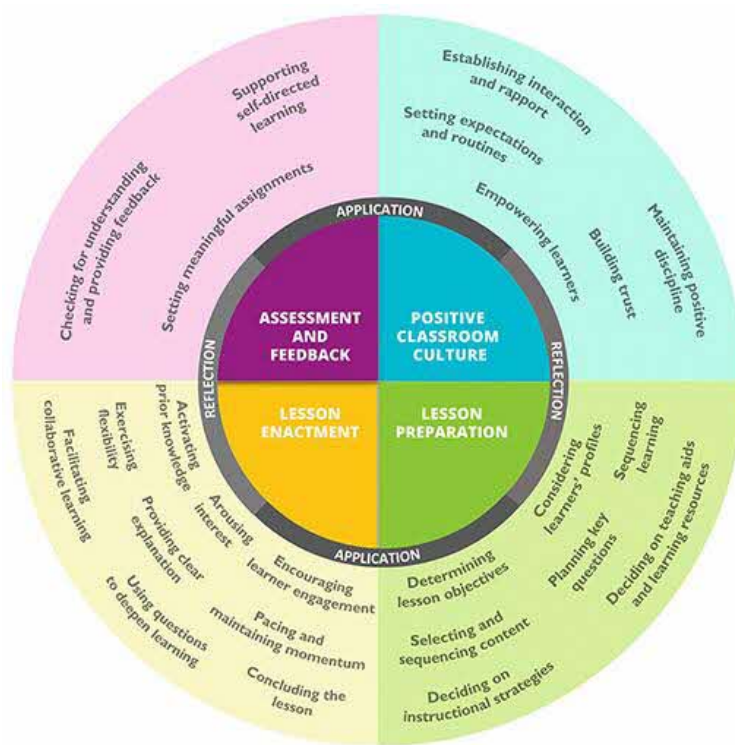


The Pedagogical Practices at the centre of the STP model are guided by the Singapore Curriculum Philosophy and supported by Knowledge Bases that provide teachers with an understanding of subject matter and goals, theories and research on students and learning, as well as theories and research on teaching.

The Pedagogical Practices (PP) from the centre of Figure 1 above are further explained in Figure 2 using the PP framework, which comprises four fundamental teaching processes that lie at the heart of good teaching: Positive Classroom Culture, Lesson Preparation, Lesson Enactment, and Assessment and Feedback (MOES, 2025). These core teaching processes make explicit what teachers carry out before, during, and after their interaction with students across various learning contexts. Through this continuous cycle of application and reflection, teachers deepen their professional competencies to enhance the quality of teaching and learning. The 24 teaching areas provide teachers with teaching actions and considerations that can be adapted to meet the students' diverse learning needs.

Figure 2

The Pedagogical Practices in the Singapore Teaching Practice



Note. The four core teaching processes and 24 teaching areas that form the PP.

These 24 teaching areas aim to sustain and enhance students' will and skill. For instance, in the teaching area of *Encouraging Learner Engagement* under the teaching process of *Lesson Enactment*, teachers focus on fostering both the will and skill of their students. Similarly, in *Setting Meaningful Assignments* under the teaching process of *Assessment and Feedback*, teachers design tasks that help students see purpose and relevance in their learning, thereby strengthening both their will and skill.

Hence, if teachers can accurately identify their students' will and skill and adopt appropriate instructional strategies accordingly, it would not only enhance learning but also support teachers in addressing their students' diverse learning needs more effectively.

Literature Review

What and Why of Will and Skill

McCombs & Marzano (1990) defined skill as developing cognitive or metacognitive competencies through training and practice, and students will refer to their desire to engage in self-regulation. To do so, students must believe that they are innovative, responsible, and independent learners who actively work towards their goals. Similarly, Zusho, Pintrich & Coppola (2003) identified students' levels of motivation and cognition in learning as will and skill, respectively, where motivation can take many forms - such as adaptive and maladaptive motivation, as described in the Motivation and Engagement Wheel (Collie & Martin, 2019; Martin, 2009).

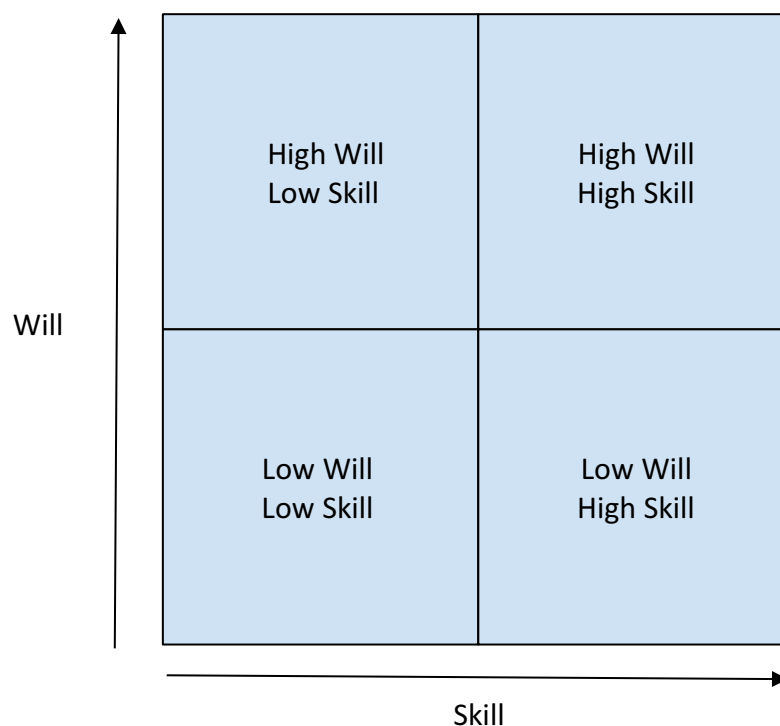
For students to experience success in the classroom, it has been argued that students must possess both the will and the skill to promote self-regulated learning and be academically adept (Alfassi, 2003; Hefter et al., 2018; McCombs & Marano, 1990; Paris & Paris, 2001; Pintrich & Groot, 1990). Wang et al. (2021) further suggested that fostering students' will, skill, and thrill can enhance learning engagement. In this educator project, skill is defined as cognitive competency, while will is defined as students' motivation towards academic learning, that is, an individual's effort, interest, and drive in relation to learning and achievement (Martin et al., 2015).

The Skill/Will Matrix

Landsberg (2015) developed the Skill/Will Matrix as a tool for leaders to engage employees, optimise performance and manage talent effectively and efficiently. In education, a similar will/skill tool was introduced for teacher professional development, in teaching with technology to enhance learner engagement and meet academic achievements (Knezek & Christensen, 2015). Likewise, Jackson (2013) presented a model that presented teaching as a combination of skill and will, aimed at supporting teachers' ongoing professional development. As shown in Figure 3, the Skill/Will Matrix found in Jackson's book categorises teachers into four types, each representing different levels of will and skill.

Figure 3

The Skill/Will Matrix



Note. The skill/will matrix identified four teacher profiles or types: high will/low skill, low will/low skill, high will/high skill, and low will/high skill.

At present, there is limited evidence of the Skill/Will Matrix being applied as a tool for student engagement. Given that all individuals possess varying levels of will and skill (McCombs & Marzano, 1990), it is possible that the Skill/Will Matrix could be adapted for use in the classroom. This possibility aligns with earlier findings that students must possess both will and skill to experience academic success (Alfassi, 2003; Hefter et al., 2018; McCombs & Marano, 1990; Paris & Paris, 2001; Pintrich & Groot, 1990).

The purpose of identifying students' level of skill and will is not to label them—labelling can have negative effects—but rather to enable educators to make informed decisions when planning and enacting lessons. Educators should also recognise that students' levels of will and skill are interrelated and may change over time. For instance, researchers found a significant decrease in motivation as students progress through their schooling, particularly among low achievers (Wang et al., 2021; Zusho et al., 2003). Accurately determining students' level of will and skill would allow educators to design lessons to effectively engage their learners.

Engagement and Motivation

Engaging students in learning is defined as the quality of their interactions with the learning tasks behaviourally, cognitively and emotionally (Wang et al., 2021). When students are engaged positively in the classroom, it helps prevent disaffection and maladaptive coping strategies such as escaping, concealing and self-pity, while promoting adaptive coping strategies such as help-seeking, strategising, and encouraging oneself (Collie & Martin, 2019). Through positive engagement, students achieve small successes that, in turn, motivate them to continue to invest in their learning (Jackson, 2011). Being engaged in lessons also enables our

students to find relevance, value, and purpose in their learning, thereby increasing their motivation and will (Skinner et al., 2016).

Motivation and engagement are closely interrelated, and both play important roles in supporting students' learning, academic achievement, and success (Collie & Martin, 2019). For this educator project, the scope of engagement will be limited to behavioural and cognitive.

Engagement in the classrooms can happen during the learning process and the demonstration of learning. Teachers often use activities such as Kahoot! to engage their students; however, many of these strategies focus on the demonstration of learning rather than the learning process itself. Engagement should begin at the very start of the learning, not only during the demonstration or assessment stage. For instance, to spark interest and nurture a lasting passion for learning across subjects—or to reignite students' curiosity—teachers should develop creative and joyful teaching experiences guided by the Relevant, Appealing, and Personal (RAP) pedagogical guidelines. Such approaches can move students' motivation from extrinsic to intrinsic, transforming their mindset from “having to learn” to “wanting to learn” (Amir, 2022; Early Childhood Development Agency [ECDA], 2023; National Institute of Education, Singapore, 2024). This switch, in turn, strengthens their will by fostering positive learning attitudes. Hence, sustaining engagement throughout the learning process ensures that assessments more accurately reflect students' understanding, thereby enhancing academic achievement and strengthening will and skill.

When selecting engagement strategies, Kagan (1989) emphasised the importance of educators being aware of the lesson objectives, understanding what these strategies are,

recognising how these strategies affect cognitive, social and emotional engagements, and knowing how and where to implement them.

Behavioural and cognitive engagement in the classroom includes instructional strategies such as differentiated instruction, cooperative learning, graphic organisers, visible thinking routines, and methods to check for understanding. These instructional strategies will be examined in this educator project for their effectiveness in engaging students in the four will and skill types.

Primary Observations

Teachers Should Know Students Well to Accurately Identify Their Level of Skill and Will

When observing in the public schools, it was observed that one of the cooperating teachers was able to easily identify the skill and will of students in three out of his six classes; however, he was unable to categorise the students in the remaining three classes, as he had only known them for three weeks. This observation suggests that to accurately identify students' level of skill and will, teachers need sufficient time to build relationships with their students and to know them well through class interactions and observations.

Use Qualitative and Quantitative Data to Identify Students' Level of Skill and Will

When identifying students' levels of will and skill, the cooperating teacher utilised both qualitative and quantitative data, such as recent test scores. Therefore, beyond knowing their students well, teachers should also use a combination and variety of qualitative and quantitative data sources to more accurately assess students' level of skill and will.

Small Successes Improve Students' Will

The cooperating teacher shared that several low-will students in his pre-algebra class performed well on their exponent quiz. Following this achievement, the teacher noticed a positive change in their learning attitudes. I concurred with the teacher's observations and noted that, during a subsequent Kahoot! activity, these students were more engaged. They took time to think through their answers before responding and, in some instances, expressed visible regret when they realised they had chosen an incorrect answer. Such positive behaviours and emotional responses indicated an improvement in their level of will. This outcome suggests that experiencing small academic successes can strengthen and sustain students' motivation, will-to-learn, and academic competence.

Presence of Low-willed Students

Teachers work hard to engage their learners. It is important to note that, regardless of how engaging a teacher may be, there might always be a few students who demonstrate a lower level of will. Jackson (2011) explained that students often face internal barriers—such as fear of failure, lack of relevance, and lack of trust—that cause them to resist learning in the classroom. Each of these barriers has different characteristics or behaviours, which can help teachers to identify the underlying reasons for students' disengagement.

Results

Characteristics and Behaviour of the Four Will and Skill Types

The characteristics and behaviours of the four types of students presented in Appendix A were collated from reading research literature (Collie & Martin, 2019; De Castella et al., 2013;

Jackson, 2011; Zusho et al., 2003), relevant books, classroom observations, and informal interviews with teachers. With these characteristics and behaviours identified, it is important to match instructional strategies with each type of learner to ensure that students are meaningfully engaged in learning, thereby improving or sustaining their levels of will and skill, as well as improving their motivation and academic success.

Effectiveness of Behavioural and Cognitive Engagement Strategies on the Will and Skill Types

Table 1 presents the effectiveness of various instructional strategies for each of the four will and skill types. The instructional strategies are briefly explained below the table, which has been compiled from personal teaching experiences, interviews and classroom observations of teachers, as well as tools shared by my Indiana University of Pennsylvania (IUP) audit professor and faculty advisor. Additional insights were gathered from relevant books, research literature and professional development websites.

In Table 1, an engagement strategy marked with an “X” indicates that that strategy is effective for a particular type of student. A strategy marked with a circle “○” indicates that the strategy needs to be modified to be effective for that student type. Strategies that are unmarked suggest limited or no effectiveness for that specific student type.

Table 1

Effectiveness of Various Instructional Strategies for Each of the Four Will and Skill Types

No.	Strategies	Student Type			
		Low Will / Low Skill	Low Will / High Skill	High Will / Low Skill	High Will / High Skill
1	Differentiated Instruction E.g. I Have, You Have, Think Dots, Cubing, Choice Boards, Error Analysis, Reversal, Parallel Tasks, Open Questions, Examples/ Non-examples, etc.	○	○	×	×
2	Graphic Organisers E.g. Frayer Model, K-W-L Chart, Flow Chart, Venn Diagram, etc.		○	×	×
3	Cooperative Learning Strategies E.g. Jigsaw, Student-Teams-Achievement Divisions (STAD), One Paper, Think-Pair-Share, etc.	○	○	○	×
4	Visible Thinking Routines E.g. See, Think, Wonder; Think, Pair, Share; I Used to Think...I Think, etc.		○	×	×
5	Checking for Understanding E.g. Muddiest Point, Most Valuable Point, Exit Tickets, Replay Signals, Gamification such as Kahoot, Gimkit, etc.		○	×	×

Note. Descriptions of these strategies, additional strategies, and suggested resources can be found in Appendix B.

Differentiated Instruction

According to Ferlazzo (2024, p.2), “Differentiated Instruction (DI) is a teaching approach that tailors instruction to address the needs of students. The instruction is varied based on the students’ learning style, strengths, and interests.” Teachers can differentiate their lessons either through content, process, product, or classroom environment. DI seeks to engage all learners and meet their diverse needs. By doing so, students' skill levels improve, and when their learning needs are met, their motivation to learn and succeed also increases; therefore, DI is effective for students who are high will/high skill and high will/low skill. However, for DI to be effective with low-will/high-skill and low-will/low-skill students, teachers need to help these students see the relevance in the learning.

Graphic Organisers

Graphic organisers are learning tools that help students to organise and make sense of their learning, for example, through the use of a Frayer model, flow chart, K-W-L chart, Venn diagram. When used effectively, graphic organisers can improve students’ understanding and knowledge retention, as well as identify learning gaps and misconceptions (Ponce 2018; Wise & Cooper, 2019).

According to Perrotta & Bohan (2013), graphic organisers alone do not improve student engagement. However, Ponce (2018) found that when graphic organisers are used during the learning process and scaffolded with essential questions, they can increase student participation, improve retention of learning and enhance recall of factual information.

Graphic organisers are effective for high-will/high-skill students. Some modifications may be needed for high-will/low-skill students to ensure relevance. For instance, teachers

might use graphic organisers during revision sessions prior to quizzes or as weighted assessments. For low-will/low-skill students, graphic organisers may be less effective due to those students' lack of motivation and lower cognitive abilities.

Cooperative Learning Strategies

Peer support and relationships play a crucial role in the growth and development of students. Positive peer support and relationships have been shown to strengthen students' engagement in learning, academic motivation, and resilience (He et al., 2025). Leveraging peer relationships through cooperative learning can improve both the skills and motivation levels of individuals within a group (Kagan, 1989; Slavin, 2014). This instructional strategy can strengthen both will and skill of low-willed students (Slavin, 2014).

To increase student achievements, Slavin (1977, 1988) discovered that two conditions must be met. First, the group must share a common goal that is important to everyone, and second, the group's success should depend on the individual performance of all its members. Ryan & Patrick (2001) found that students generally did not become more disruptive when encouraged to engage in academic conversations with one another. Therefore, teachers should not hesitate to facilitate cooperative learning due to classroom management concerns.

Educators should thus consider using cooperative learning strategies to engage students with low will or low skill. Cooperative learning strategies are effective for students who are high will/high skill. Low-will/high-skill students who may exhibit conflicting behaviours such as overconfidence or disengagement, require careful facilitation to remain engaged. For low-will/low-skill students, cooperative learning strategies should be modified — for instance, by grouping them with inclusive and familiar peers- to create a positive, safe, and supportive

environment that encourages participation and learning. For high-will/low-skill students, grouping them with peers who have higher skill levels can provide academic support. When cooperative learning involves pair work, pairing students with low skill levels should be avoided. Asking low-skill students to work together may limit learning outcomes and reduce engagement. Additionally, assigning group roles that leverage the strengths of these high will/low skill students can also boost self-confidence and self-esteem, thereby strengthening their will and skill.

Visible Thinking Routines

Visible thinking routines such as those presented in *Making Thinking Visible* (Ritchart & Perkins, 2008) are engagement tools. According to Harvard University's Project Zero (n.d.), there are ten types of thinking routines, each serving different purposes. For example, Chalk Talk is used to introduce and explore ideas, I Used to Think...now I Think... helps students synthesize and organise ideas, and Think, Pair, Share encourages students to consider controversies, dilemmas and perspectives.

When teachers facilitate visible thinking routines, students develop mental processes such as solving, decision-making, clarifying, planning, organising, and creating new ideas (Moore, 2015). Moreover, the classroom becomes learner-centred and all students' ideas are valued, allowing teachers to accurately assess students' understanding and improve learning outcomes, confidence and achievements (Ritchart & Perkins, 2008). Gholam (2018) further highlighted that thinking routines enhance student engagement, reinforce thinking skills and develop 21st-century competencies.

Visible thinking routines are most effective to engage high-willed students. Low-willed students will not benefit from this strategy because it requires sharing their thinking, which they often avoid to prevent being perceived as failures. For low-will/high-skill students, visible thinking routines can be paired with cooperative learning strategies or other engagement tools. For instance, a teacher might conduct thinking routines in a one-to-one setting to facilitate students' participation while making their thinking visible.

Formative Assessments to Check for Understanding

Formative assessments, such as traffic lights, exit tickets, and replay signals, are strategies that teachers use to check for student understanding (Leonard, 2024). Similar to visible thinking routines, consistently using formative assessments rather than simply asking "Are there any questions?" allows teachers to gauge students' level of understanding more accurately. This student-centred approach enhances students' engagement levels, learning and academic achievements. High-willed students would benefit from formative assessments, while the effectiveness for low-willed students depends on how the teachers implement formative assessments.

It should be noted that Table 1 is not a comprehensive list of instructional strategies. Appendix B provides additional strategies, their descriptions, and suggested resources. More strategies can be added following further investigation.

Discussion

Table 2 shows instructional strategies that are suitable for high-will/high-skill students and are fairly effective for high-will/low-skill students. These strategies would require

modifications to engage the low-will/high-skill students due to their complacent attitudes.

Notably, there are relatively few strategies that effectively engage low-will/low-skill students.

The following discussion focuses on additional modifications or remedial approaches to support the engagement of low-willed students more effectively.

Modifying Instructional Learning Strategies for Low-Will/Low-Skill and Low-Will/High-Skill Students

Jackson (2013) suggests that instructional coaches deploy listening skills to understand what a low-skill/low-will teacher says and to observe actions made. In the context of this educator project, teachers are encouraged to use a similar approach to understand their low-will/low-skill students - determining whether disengagement is primarily a skill or a will issue - and respond or act accordingly (Jackson, 2013).

If disengagement is primarily a skill issue, teachers should focus on closing the learning gap promptly to enhance students' skill level, learning engagement, and motivation and will. On the other hand, if the root cause of disengagement is a will issue, Jackson (2011, p.17) found that students' resistance towards learning often arises from fear of failure, lack of relevance, or lack of trust. To address these issues, educators should avoid using generic strategies; instead, they should adopt a positive classroom culture and instructional strategies that address the underlying cause of disengagement. As the purpose of this educator project is to identify effective instructional strategies for engaging the four will and skill types, Table 3 outlines specific instructional strategies that teachers can implement once the root cause of low will has been identified.

Table 3

Specific Instructional Strategies to Address the Primary Cause of Low Will

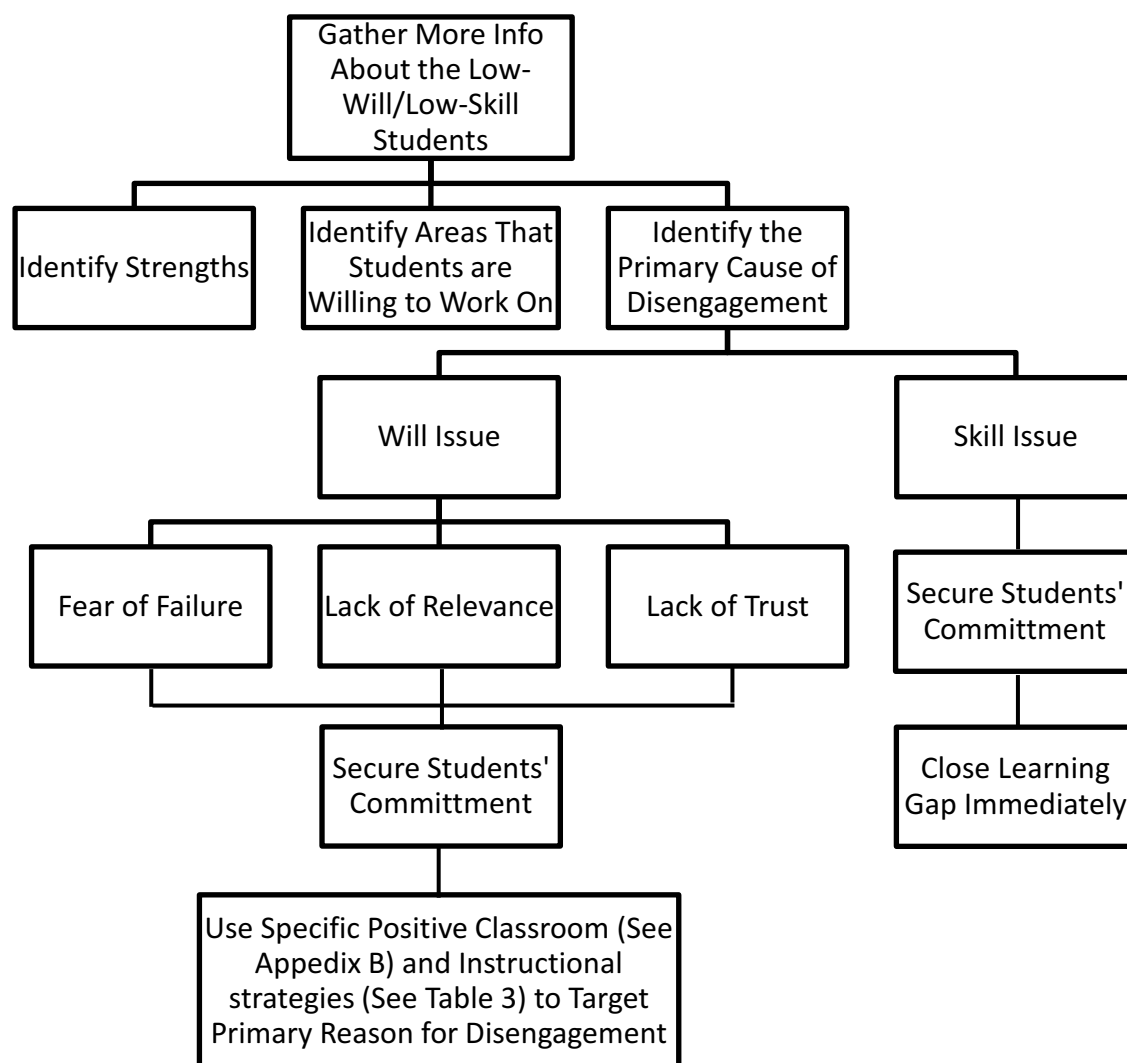
Reasons for Resisting Learning	Instructional Strategies to Try
Fear of Failure.	● Ask for choral responses. ● Use cooperative pairs. ● Use numbered heads ● Have students assume roles for discussion. ● Use error analysis.
Lack of Relevance	● Use advance organisers. ● Choose culturally relevant materials. ● Create study groups. ● Build background knowledge. ● Use the statement strategy. ● Use anticipation guides for texts.
Lack of Trust	● Use cooperative learning strategies ● Use check-for-understanding strategies ● Use examples/non-examples ● Use open questions

Note. Currencies are knowledge, soft skills, social skills, and network affiliations.

Before the implementation of remediation, teachers should seek to secure students' commitment, provide clear and specific directions, and offer targeted guidance to steer students towards success. Figure 4 presents a flowchart that educators can use as a guide when supporting low-will/low-skill students.

Figure 4

A Guide When Supporting Low-Will/Low-Skill Students



By adopting instructional strategies that address the primary cause of low will, teachers not only engage students with low will/low skill but also enhance the motivation of those with low will but high skill, thereby increasing their learning engagement and strengthening their overall will.

Other Ways to Improve or Sustain Students' Will or Skill

In addition to modifying instructional strategies that support students with low will, Jackson (2011) recommends cultivating a positive classroom culture through specific strategies that address the root cause of disengagement. These strategies can be found in Appendix C.

To develop the will and skill of low-will/low-skill students, Alfassi (2003) suggests that schools implement a learner-centred and structured academic program that fosters academic competence and self-efficacy. Such programs have been shown to significantly improve students' achievement and self-efficacy, as well as improve their internal motivational orientation. Intentionally creating success can raise self-efficacy, while repeated failure can lower it.

Wang et al. (2021) found that although students' engagement in mathematics declined over time, those with stronger metacognitive skills, greater interest and higher self-control reported, on average, a higher daily engagement. Although Wang et al. conducted their research in mathematics, their research could be applied to any subject. Hence, schools should focus on developing students' metacognition skills and self-control, while also fostering situational interest to increase effective learning. These three constructs—metacognition, self-control, and interest—can work together mutually or counterbalance one another to support engagement, motivation, and will and skill.

Schools should also promote adaptive motivation and engagement while reducing maladaptive ones, for instance, by creating a classroom environment that encourages students to appraise themselves. This practice allows students to feel more capable, increases their self-

efficacy, and their expectations for success and self-concepts. Overtime, students develop adaptive goals and greater willingness to commit to their learning (Collie & Martin, 2019).

Anxious students are less likely to put in effort to learn in order to avoid making mistakes (Ryan & Patrick, 2001). To address this challenge, schools should create a classroom environment where students' ideas and efforts are respected with minimal threat of those students being embarrassed or teased, and where mistakes are part of the learning process. Such a learning environment boosts students' confidence and encourages them to devote more cognitive resources to engage and learn. Growth mindset training can support this aim and help students embrace challenges.

Teachers should also be understanding and available to help students without judgment. When students perceive that their teachers are understanding and helpful, they display fewer off-task and disruptive behaviours (Ryan & Patrick, 2001). To sustain students' high will and high skill, teachers can conduct a variety of classroom activities aligned with the six categories of Bloom's taxonomy. Gradually increasing cognitive challenges encourages engagement, deepens understanding, and promotes gradual cognitive development and skill (Oosthuizen, 2019; Shneiderman, 1998; Sidekerskienė, 2025; Tay, 2025;).

Limitations

This paper is an educator project rather than a research project. As such, there remains significant potential for in-depth research on the characteristics and behaviour of the four types of learners based on their levels of will and skill, as well as on the effectiveness of the various instructional strategies used to engage each type. This project also examined a limited range of instructional strategies. There is potential, therefore, for further investigation into the

effectiveness of other strategies that were not included in this study, such as harnessing brain-based principles.

The Skill/Will Matrix may risk categorising students too rigidly into one of the four types. Therefore, there is potential to explore a scaled model that allows for greater flexibility and recognises the continuum of learners' will and skill levels.

Educator Project Implementation & Conclusion

This educator project identified the different characteristics and behaviours of students with different levels of will and skill. It also compiled various instructional strategies commonly used by teachers to engage students and examined their effectiveness across the four will and skill types.

As the Skill/Will matrix has not been previously applied directly to students, there is potential to enhance further the accuracy and application of the findings from this project.

Upon returning to Singapore, I plan to do the following:

- Share the findings of this educator project with fellow educators to raise awareness, gather feedback, and promote an exchange of dialogue.
- Continue to collect data and enhance the findings by implementing these instructional strategies in my own classes. I also will continue to collect data to strengthen the validity of the findings in this project.
- Collect data from teachers in my school on the characteristics and behaviours of the four will and skill types, as well as their perception of the effectiveness of various instructional strategies for each of the four will and skill types.

- Collaborate with interested teachers to refine and enhance the results presented in this project.
- Identify students representing each of the four will and skill types and collect quantitative data, such as academic results, and qualitative data through interviews and classroom observation.
- Share the research findings with teachers from other schools to contribute to professional development.

Through these efforts, this project aims to deepen the understanding of how teachers can better engage students of differing will and skill levels, thereby supporting their motivation, learning, and academic success. In turn, students may receive more effective and timely support that strengthens their will and skills in academic motivation and competence.

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Appendices

Appendix A

Typical Characteristics and Possible Actions and Behaviours of the Four Will and Skill Types

Student Type	Typical Characteristics	Possible Actions and Behaviours
Low Will / Low Skill	• Disengaged • Discouraged • Dejected • Not invested • Accepting of failure • Low in success orientation and failure evasion. • Low self-esteem • Low control belief • Indifference • Loss of hope • Learned helplessness	• Does not volunteer or contribute. Appears passive • Avoids the teacher's gaze when a question is asked • Does not bring any learning materials • Looks out the window / Sleeps in class / Puts head on the table • Refuses to try / Gives up easily even with support • Does not take up the support teachers have put in place • Does not want to know how "easy" or "hard" a task is, as they do not even want to attempt it • Answers but without thinking/trying • Does not submit work or submits incomplete work • Tends to procrastinate • The more the teacher pushes, the harder the student resists • Does not care about what the teacher says about their performance • Does not care about grades at all • Indifferent to success and failure / Seems to expect to fail • Does not care if they are in situations where they might look "stupid". • Indicates little shame at their failures / Takes little pride in their success • Likely to get zero for any assessment • Is oppositional, defiant or angry • Refuses to follow class rules and procedures • Takes a long time to get started on a task • Withdraws from learning or school

Student Type	Typical Characteristics	Possible Actions and Behaviours
High Will / Low Skill	• Enthusiastic • Idealistic • Willing to learn • Low self-efficacy • High perceived control • Fear of failure • High success orientation • Hardworking and persistent • Anxious	• Attempts to plan, albeit poorly • Attempts work, but mostly gives wrong responses • Pays attention in class, but may have misconceptions in their learning • Consistently looks for help (peers or teachers) • Displays poor comprehension skills • Blames self for not doing well • Remains quiet/hidden during lesson • Seen as trying, but could be struggling in silence • Distracted by personal belongings. • Relies heavily on others for input. Adopts adaptive positive engagement
High Will / High Skill	• Persistent • Motivated • Skillful • Optimistic • Self-confident • Resilient • Proactive orientation to tasks • High and positive self-efficacy • High self-worth • High perceived control	• Always on task • Makes plans • Manages tasks well • Adopts mastery orientation • Adopts adaptive positive motivation and engagement • Values self • Submits quality work • Displays good study skills • Takes down notes • Demonstrates active listening • Seeks extra consultation • Participates actively in class (e.g. first few to answer questions posed by teachers) • Refers to notes to recall before attempting the task • When not engaged, finds other things to do to occupy their time • Maybe easily distracted, with the ability to be on task quickly when prompted • Displays exemplary achievement behaviours

Student Type	Typical Characteristics	Possible Actions and Behaviours
		•
Low Will / High Skill	• Skillful but disinterested • “Seen it all” • Complacent • Overconfident • Low in success orientation and high in failure evasion • Self-Protective • Vulnerable to self-handicap • Defensive • Pessimistic • Self-handicapping • Learned helplessness	• Shows signs of anxiety nearing test dates • Avoids situations where they might look “stupid” • Looks for shortcuts/asks “Will this be on the test?” • Engages in distracting behaviour • Does minimal work by rushing through assignments as quickly as possible without much thought into it • Wears a mask of confidence • Is very grade-conscious • Focuses on the grade rather than the learning • Is oppositional, defiant or angry • Intentionally annoys people with authority • Blames the teacher or other external factors for their poor grades (i.e. engages in self-protective behaviour) • Refuses to follow class rules and procedures • Ridicules and shows disrespect for authorities • May multitask with other tasks which are more engaging to them • Frequently comments about how easy the work is • Complains that class is boring • Late for class and among the first few to pack early and leave the classroom • Instead of getting ready for the lesson, uses the time to engage in other activities (e.g. to play on their phone) • Attempts to delay the start of class (e.g. asks for more time to finish a game, asks to go to the toilet upon class starting) • Plays truant

Appendix B

Compilation of all strategies, brief description, and resources of these engagement strategies shared in Table 1

Strategies to Differentiated Instruction	Brief Description	Resources
<i>I Have, Who Has / I Have, You Have</i>	<i>The “I Have, Who Has” teaching method is an interactive, game-based strategy in which each student holds a piece of information. One student begins by reading their information aloud and then calls on another student to respond with the information they hold. For example: “I have 9 a.m., who has 10:35 a.m.?” This method promotes active participation and quick recall, making learning both engaging and collaborative.</i> <i>Alternatively, the strategy into an “I Have, You Have” interactive activity, where one student begins by reading their information aloud and then reads the information presented by a classmate. This variation encourages active participation and reinforces the application of learning.</i>	<i><u>“I Have, Who Has” from The Mountain Teacher</u></i>
Think Dots	<i>Think Dots is a strategy that provides students with six different activities, each assigned a value from one to six dots. It is designed to help students review, demonstrate, and extend their thinking. While students are given a degree of choice, they are still required to complete specific learning tasks—either by rolling a die to determine which task to complete or by selecting a predetermined number of tasks out of the six options. Variations of this strategy can also be implemented, such as requiring students to complete enough activities to accumulate a specific number of dots as determined by the teacher.</i>	<u>Cubing & Think Dots by Eulouise Williams</u> <u>Think Dots by Openingpaths.org</u>

Strategies to Differentiated Instruction	Brief Description	Resources
Tic-Tac-Toe Boards	<i>Think-Tac-Toe</i> is an instructional strategy that uses a grid of nine squares, each containing a different task. Students select and complete a row, column, or diagonal of tasks, allowing for both choice and differentiation in learning.	Think-tac-toe by Thoughtco Think-Tac-Toe by TeachLikeThis Think-tac-toe by Blue Mountain Math
Choice Boards / Learning Menus	<i>Choice Boards</i> or <i>Learning Menus</i> are virtual or printed learning tools that provide students with a range of activities and scaffolds, allowing them to choose, complete, and demonstrate their learning independently.	Choice Boards by Edutopia Choice Boards by Brenda Iasevoli, HMH Choice Boards by Dr. Catlin Tucker
Cubing	In <i>Cubing</i> , students use a cube with a different activity on each face, allowing them to explore a concept from multiple perspectives. A student rolls the cube and completes the activity that appears on the top face. Students with lower skill levels can begin with cubes containing more basic activities to help build their skills and confidence before progressing to more complex tasks.	Cubing & Think Dots by Eulouise Williams
Error Analysis	In <i>Error Analysis</i> , the teacher provides students with incorrect solutions, and students are tasked with identifying and correcting the errors.	Error Analysis by Edutopia
Super Wrong, Kinda Right	In <i>Super Wrong, Kinda Right</i> , the teacher asks students to create incorrect answers or solutions that range from being completely	Super Wrong, Kinda Right by Instructional Nudges

Strategies to Differentiated Instruction	Brief Description	Resources
	wrong to almost correct. Students are then required to explain their reasoning behind each response, encouraging them to analyse misconceptions and deepen their conceptual understanding.	
Analyze This	In <i>Analyze This</i> , teachers provide students with two different representations of the same problem and ask them to compare and analyse these representations.	Analyze This by Instructional Nudges
Pairing	In <i>Pairing</i> , students select a pair of problems they believe are the most similar or the most different, and then explain the reasoning behind their choices.	Pairing by Instructional Nudges
Reversal / Turning Around a Question	In <i>Reversal</i> or <i>Turning Around a Question</i> , students are provided with the solution and are asked to create a new problem based on it. This strategy promotes flexible thinking and gives students additional practice to generalise their understanding independently, without placing extra demands on the teacher to create exercises.	Reversal by Instructional Nudges Reversals by Two-Minute Teacher's Guide
Mathematics Open Middle Problems	<i>Open Middle Problems</i> are problems that begin with the same initial conditions and have the same final answer, but allow multiple approaches and solution strategies. This type of problem encourages students to develop a deeper conceptual understanding of mathematics.	Open Middle Problems by Open Middle
Mathematics Numberless Problems	A <i>Numberless Word Problem</i> is a problem from which all numerical values have been removed. This approach fundamentally changes the way students engage with word problems, encouraging them to focus on understanding the context and reasoning behind the problem rather than rushing into calculations. It helps reduce calculation-	Numberless Word Problem by Brian Bushart

<i>Strategies to Differentiated Instruction</i>	<i>Brief Description</i>	<i>Resources</i>
	driven behaviour and promotes deeper conceptual comprehension.	
Examples/ Non-examples	In <i>Examples/Non-Examples</i> , students create at least two examples and two non-examples of a given term, concept, or procedure. They are also encouraged to include examples about which they are unsure. This strategy helps students demonstrate conceptual understanding and reduces the likelihood of misconceptions.	Examples / Non-examples by Instructional Nudges
Open Questions	<i>Open Questions</i> are questions that allow for a variety of responses or approaches. Strategies for creating open questions include: • Turning around a question • Asking for similarities and differences • Replacing a number with a blank • Creating a sentence • Using “soft” words • Changing the question These strategies encourage flexible thinking, deeper reasoning, and multiple approaches to problem-solving.	Book- Good Questions: Great Ways to Differentiate Mathematics Instruction in the Standards-based Classroom Open Questions and Parallel Tasks by Powerful Learning
Parallel Tasks	<i>Parallel Tasks</i> are a set of two to three tasks designed to meet the needs of students at different levels of learning. All tasks focus on the same knowledge, concept, skill, or big idea, but are sufficiently similar in context to allow all students to engage meaningfully.	Book- Good Questions: Great Ways to Differentiate Mathematics Instruction in the Standards-based Classroom Open Questions and Parallel Tasks by Powerful Learning

Cooperative Learning Strategies	Brief Description	Resources
Think, Pair, Share	<i>Think-Pair-Share</i> (TPS) is a collaborative problem-solving model developed by Professor Frank T. Lyman Jr. and educator Arlene Mindus. It follows a three-step sequence: first, students take time to think about questions posed by the teacher; next, they pair up with a partner to share their answers, discuss, and refine their responses; finally, each pair shares their insights with the rest of the class.	16 variations of Think, Pair, Share: by Edutopia Think-pair-Share by Project Zero, Harvard Graduate School of Education
Chalk Talk	<i>Chalk Talk</i> is a collaborative learning strategy and thinking routine in which students silently reflect on ideas, questions, or problems by writing their thoughts. They also respond to the contributions of others, promoting thoughtful discussion and idea development.	Chalk Talk by Project Zero, Harvard Graduate School of Education
Jigsaw	The <i>Jigsaw</i> teaching method is a cooperative learning strategy in which students are divided into groups, with each member assigned a different portion of the material to learn. Students then meet with peers from other groups who have studied the same portion to become “experts” before returning to teach their original group members. This approach fosters active learning, accountability, and teamwork, as each student’s contribution is essential to completing the “whole picture” of the lesson.	Jigsaw by the Jigsaw Classroom
Single/Continuous/ All Write/ Timed Round Robin	<i>Single/Continuous/Timed Round Robin</i> are cooperative learning strategies where students are grouped together, and each student takes turns sharing their answers with their groupmates. This cooperative learning strategy can occur in a single round, continuously, or for a set time determined by the teacher. In <i>All Write Round Robin</i> , students write their answers individually on their own paper rather than sharing them orally.	Kagan Cooperative Learning by Kagan Publishing & Professional Development Kagan Structures by San Felipe-Del Rio Consolidated Independent School District, Texas

Cooperative Learning Strategies	Brief Description	Resources
Rally Robin	<i>Rally Robin</i> is a cooperative learning strategy in which students work in pairs and take turns sharing one answer or brainstorming an idea.	Kagan Cooperative Learning by Kagan Publishing & Professional Development Kagan Structures by San Felipe-Del Rio Consolidated Independent School District, Texas
Give One, Get One	In the <i>Give One, Get One</i> cooperative learning strategy, students work in pairs and take turns sharing one answer or brainstorming an idea. They then discuss the merits and differences of their responses. Afterward, students find a new partner and repeat the activity, allowing them to exchange and evaluate multiple ideas.	Give one, Get One by Opportunity Education
Timed, Pair, Share	<i>Timed Pair Share</i> is a cooperative learning strategy in which students pair up and share their ideas with a partner for a set period of time. During this time, the partner listens attentively. After the allotted time, the roles are switched so that both students can share and listen.	Kagan Cooperative Learning by Kagan Publishing & Professional Development Kagan Structures by San Felipe-Del Rio Consolidated Independent School District, Texas
Rally Coach	<i>Rally Coach</i> is a cooperative learning strategy in which students work in pairs and take turns. One student works on a problem while the other provides coaching and guidance. The roles are then switched so that both students can solve and coach.	Kagan Cooperative Learning by Kagan Publishing & Professional Development Kagan Structures by San

Cooperative Learning Strategies	Brief Description	Resources
		<u>Felipe-Del Rio Consolidated Independent School District, Texas</u>
Stand Up, Hand Up, Pair Up	<i>Stand Up, Hand Up, Pair Up</i> is a cooperative learning strategy in which, upon receiving instructions, students stand up and raise their hands. They then quickly find a partner who is also standing with their hand up and begin their discussion or sharing activity.	<u>Kagan Cooperative Learning by Kagan Publishing & Professional Development</u> <u>Kagan Structures by San Felipe-Del Rio Consolidated Independent School District, Texas</u>
Numbered Heads Together	<i>Numbered Heads Together</i> is a cooperative learning strategy in which students work in groups, with each member assigned a number. The group discusses the answers together, and the teacher then calls out a number. The student with that number from each group stands and responds simultaneously, ensuring accountability and participation from all group members.	<u>Numberer Heads Together by Be GLAD!</u> <u>Kagan Structures by San Felipe-Del Rio Consolidated Independent School District, Texas</u>
Pairs Check	<i>Pairs Check</i> is a cooperative learning strategy in which students work in pairs within groups of four. Within each pair, students carry out the <i>Rally Coach</i> strategy. After completing every two problems, the pairs check their	<u>Pairs Check by Kagan Publishing & Professional Development</u>

Cooperative Learning Strategies	Brief Description	Resources
	answers with the other pair in the group, promoting collaboration and accuracy.	
Three-Step Interview	The <i>Three-Step Interview</i> is a cooperative learning strategy in which students form groups of three, with each member assigned a role: interviewer, interviewee, or reporter. Students take turns performing each role, and the reporter then shares the key details gathered with the entire group, promoting active listening, communication, and collaborative learning.	Three Step Interview by LEARNstyle LearnTube
Team Word-Webbing	<i>Team Word-Webbing</i> is a cooperative learning strategy in which students simultaneously write on a shared piece of paper, identifying main concepts, supporting elements, and illustrating the connections and relationships between them. This approach promotes collaboration, organisation of ideas, and a deeper understanding of the content.	Team Word-Webbing by VicTESOL
Single/Simultaneous Roundtable	<i>Single/Simultaneous Roundtable</i> is a cooperative learning strategy in which students work independently/in groups and take turns writing their answers on a shared sheet of paper. After writing, the paper is passed to the next team member, who continues the work. This process continues until the teacher signals to stop, promoting collaboration, idea sharing, and collective problem-solving.	Kagan Structures by San Felipe-Del Rio Consolidated Independent School District, Texas Simultaneous Round Table by District 62 Media
Inside-Outside Circle	<i>Inside-Outside Circle</i> is a cooperative learning strategy in which the class is divided into two groups, each forming a circle. One group forms an inside circle facing outward, while the other group forms an outside circle facing inward. Students discuss with the partner directly opposite them, promoting dialogue, idea exchange, and active participation. To extend the	Inside-Outside Circle by Ellevation Education

Cooperative Learning Strategies	Brief Description	Resources
	learning, students in the outside circle rotate to engage with a new inside partner, allowing for broader interaction and multiple perspectives.	
Fishbowl	<i>Fishbowl</i> is a cooperative learning strategy in which the class is divided into two groups, each forming a circle. One group forms an inner circle facing inward, while the other group forms an outer circle facing the inner circle. Students in the inner circle engage in discussion, while those in the outer circle observe and practice active listening, promoting reflection, critical thinking, and engagement.	Fishbowl Discussion by Edutopia
Hot Seat	<i>Hot Seat</i> is a cooperative learning strategy in which one student assumes the role of a character while the rest of the class interviews them. The student in the “hot seat” responds to questions from the perspective of the assigned character, promoting role-based understanding, perspective-taking, and active engagement.	Hotseat by Redesign
Fishbowl With Hot Seat	<i>Fishbowl with Hot Seat</i> is a cooperative learning strategy in which students in the inner circle discuss a topic provided by the teacher, while students in the outer circle observe and practice active listening. Outer circle students then join the inner circle in the “hot seat” to share additional perspectives before returning to their original positions. This strategy promotes enriched learning, multiple viewpoints, and deeper engagement.	Fish Bowl with Hot Seat by Edutopia
De Bono’s Six Thinking Hats	<i>De Bono’s Six Thinking Hats</i> is a cooperative learning strategy that assigns specific thinking roles to students, encouraging them to investigate a problem from multiple perspectives. This strategy can also be applied individually. The six hats are:	De Bono's Six Thinking Hats by The de Bono Group

Cooperative Learning Strategies	Brief Description	Resources
	● White Hat: Focus on facts and information. ● Yellow Hat: Explore the positive aspects. ● Black Hat: Examine potential problems or risks (devil's advocate). ● Red Hat: Express feelings, intuitions, likes, dislikes, and concerns. ● Green Hat: Think creatively, generating new ideas and alternatives. ● Blue Hat: Act as the manager, ensuring all hats stay on task and the process remains organised. This strategy promotes critical thinking, perspective-taking, and structured problem-solving.	
Student-Teams-Achievement-Divisions (STAD)	<i>STAD</i> (Student Teams-Achievement Divisions) is a cooperative learning method in which students are grouped into teams of four, with diversity in performance level, gender, and ethnicity. The teacher delivers the lesson, and students work together to ensure all team members understand and master the material. Students then take individual quizzes on the content, and points are awarded based on quiz results and prior performance. These points are totalled to form team scores, and teams that meet the teacher's criteria receive rewards. This approach promotes collaboration, accountability, and mastery learning.	STAD by Robert E. Slavin
Teams-Games-Tournament (TGT)	<i>TGT</i> (Teams-Games-Tournament) is a cooperative learning method in which students are grouped into teams of four to five members with mixed performance levels. The teacher delivers the lesson, and students work together to help one another prepare. Teams then compete against two to three members from other teams in a tournament-style activity, earning points that contribute to their overall team scores. This approach promotes collaboration, peer teaching, and healthy competition.	TGT by steemit

Cooperative Learning Strategies	Brief Description	Resources
Group Investigation	<i>Group Investigation</i> is a cooperative learning strategy in which students form heterogeneous groups. Each group is assigned a topic, theme, or problem, which they plan and subdivide into tasks among members. Students then conduct investigations using various methods and sources to examine, explore, and understand the topic. Finally, each group prepares and presents their findings, promoting collaboration, critical thinking, and independent inquiry.	Group Investigation by Byrdseed
Gallery Walk	<i>Gallery Walk</i> is a cooperative learning strategy that can be conducted in various ways. One common approach involves placing flip chart papers around the classroom, each containing a different task or question. Students are organised into groups, typically based on the number of questions or stations. Each group spends a set amount of time at each station to answer the question, complete the task, or respond to prior entries on the paper. Groups rotate through all stations, ultimately returning to their starting point, and then present their findings to the entire class. This strategy promotes collaboration, critical thinking, and collective reflection.	Gallery Walks by Edutopia Gallery Walk by Facing History & Ourselves
One Paper	<i>One Paper</i> is a cooperative learning strategy in which students work together on a single piece of paper, using different coloured pens to indicate each member's contribution to the answers.	One Paper by Instructional Nudges
Snowball Toss	<i>Snowball Toss</i> is a cooperative learning strategy in which students write their responses to a question or task on individual pieces of paper. They then roll the papers into "snowballs" and toss them into the centre of the classroom. Each student randomly selects a snowball, unfolds it, and reads the response. The content can then be shared and discussed with the entire class, promoting engagement, reflection, and collaborative learning.	Snowball Toss by Edutopia

Graphic Organisers	Brief Description	Resources
Fraye Model	The <i>Fraye Model</i> is a graphic organiser designed to help students explore vocabulary in depth, understand word meanings, and build connections between concepts. By systematically examining definitions, characteristics, examples, and non-examples, the model enhances comprehension and retention across all subject areas.	Fraye Model by Teach Hub.com
K-W-L Chart	A <i>K-W-L Chart</i> is a graphic organiser that helps students connect and organise information. It consists of three columns: “K” for what students Know , “W” for what they Want to know , and “L” for what they Learn .	20 Types of Graphic Organisers by Creately
Double Bubble Map	A <i>Double Bubble Map</i> is a graphic organiser used to compare and contrast two topics, highlighting their similarities and differences.	
Venn Diagram	A <i>Venn Diagram</i> is a graphic organiser used to compare and contrast multiple topics, visually highlighting similarities and differences.	
Persuasion Map	A <i>Persuasion Map</i> is a graphic organiser that helps students plan their persuasive writing in steps, visualise how their points are connected, and identify any gaps in their argument.	
Sequence Chart	A <i>Sequence Chart</i> is a graphic organiser used to arrange events, steps, or ideas in a specific, logical order.	
Story Map	A <i>Story Map</i> is a graphic organiser that helps students break down and analyse the key elements of a story, enhancing comprehension and understanding.	
Biography Graphic Organiser	A <i>Biography Graphic Organiser</i> is a tool that helps students understand and analyse the life of a person or character, highlighting key events, experiences, and influences.	

Graphic Organisers	Brief Description	Resources
Analogy Graphic Organiser	An <i>Analogy Graphic Organiser</i> is a tool that helps students compare and contrast new concepts by relating them to something they already know, promoting understanding through familiar connections.	20 Types of Graphic Organisers by Creately
Vocabulary Graphic Organiser	A <i>Vocabulary Graphic Organiser</i> is a tool for teaching students new words by breaking down their meanings and exploring elements such as definitions, synonyms, and usage in context.	
Timeline Graphic Organiser	A <i>Timeline Graphic Organiser</i> helps students organise information into two categories. It can also be adapted to show a sequence of events in chronological order, making it useful for studying history or planning timelines.	
T Chart	A <i>T-Chart</i> is a simple graphic organiser that helps students compare and contrast two opposing ideas side by side.	
Hierarchy Chart	A <i>Hierarchy Chart</i> is a graphic organiser that helps students break down complex topics into different levels of information, illustrating relationships from general concepts to specific details.	
Star Diagram	A <i>Star Diagram</i> is a graphic organiser that helps students visualise the connections and relationships between different characteristics of a topic.	
Cluster Diagram	A <i>Cluster Diagram</i> is a graphic organiser used to visually organise ideas generated during a brainstorming session, showing relationships and connections among concepts.	
Lotus Diagram	A <i>Lotus Diagram</i> is a graphic organiser that breaks down complex topics into smaller, more manageable sub-topics, often using ideas generated during a brainstorming session.	

Graphic Organisers	Brief Description	Resources
Cause and Effect Graphic Organiser	A <i>Cause-and-Effect Graphic Organiser</i> helps students visually represent the relationship between events and their outcomes. This tool is particularly useful for analysing the causes of specific events and understanding their effects.	20 Types of Graphic Organisers by Creately
Math Graphic Organiser	A <i>Math Graphic Organiser</i> is a tool that helps students break down and organise mathematical concepts, enhancing understanding and problem-solving skills.	
Flowchart / Procedural Notes	<i>Flowcharts</i> or <i>Procedural Notes</i> are graphic organisers that model the sequence of steps in a procedure. They can be used to illustrate sample problems or guide the problem-solving process for specific concepts or topics.	Flowchart by UCL
Mind Map / Spider Map	A <i>Mind Map</i> is a graphic organiser that helps students visually connect ideas and concepts around a central topic.	Spider Diagram by Creately 20 Types of Graphic Organizers by Creately Free Online Mind Mapping by MindMup Online Mind Mapping Tool by University of Arizona Libraries
Fist List	A <i>Fist List</i> is a graphic organiser shaped like a hand, used to connect ideas around a central topic. The topic is written in the centre of the palm, and	Fist List by The Niemans' Nook, TPT

Graphic Organisers	Brief Description	Resources
	students write five ideas they remember, one on each finger. Students then share their ideas with their group or with the entire class, promoting recall, discussion, and collaborative learning.	

Visible Thinking Routines	Brief Description	Resources
See, Think, Wonder	<i>See, Think, Wonder</i> is a visible thinking routine that encourages students to make careful observations and thoughtful interpretations. It fosters curiosity and prepares students for inquiry-based learning. Guiding questions may include: <i>What do you see? What do you think about that? What does it make you wonder?</i>	See, Think, Wonder by Project Zero, Harvard Graduate School of Education
I Used to Think... Now I think	<i>I Used to Think... Now I Think</i> is a visible thinking routine that helps students synthesize, reflect on, and organize their evolving ideas.	I Used to Think... Now I Think... by Project Zero, Harvard Graduate School of Education
Connect, Extend, Challenge	<i>Connect, Extend, Challenge</i> is a visible thinking routine that helps students draw connections between new ideas and their prior knowledge. Guiding questions include: <i>What does this connect to that you already know? What new ideas extend your thinking? What is challenging or confusing? What do you wonder about?</i> This routine promotes reflection, critical thinking, and deeper understanding.	Connect, Extend, Challenge by Project Zero, Harvard Graduate School of Education
Surprising, Interesting, Troubling (SIT)	<i>Surprising, Interesting, Troubling (SIT)</i> is a visible thinking routine in which students identify and record what they find surprising, interesting, or troubling about a topic. This strategy encourages reflection, critical thinking, and deeper engagement with the content.	SIT by Facing History & Ourselves

Visible Thinking Routines	Brief Description	Resources
Other Visible Thinking Routines	Refer to resources for other visible thinking routines.	PZ's Thinking Routines Toolbox by Project Zero, Harvard Graduate School of Education

Strategies to Check for Understanding	Brief Description	Resources
Entry / Exit Ticket	Entry/Exit Tickets are formative assessment tools used to gauge students' prior knowledge at the beginning of a lesson or their understanding at the end of a lesson.	Exit Tickets by Edutopia
Replay Signals	<i>Replay Signals</i> are pre-arranged hand gestures or signals that students use to indicate to the teacher that they need a concept or explanation repeated or clarified. Teachers can also adapt this strategy using alternative materials, such as paper cups, cards, or other visual cues.	Book - Fast and Effective Assessment: How to Reduce Your Workload and Improve Student Learning, 2018, p.64-98
Rate and Review	In <i>Rate and Review</i> , teachers provide students with a sample answer or solution. Students then evaluate the response by assigning a rating and explaining the reasoning behind their evaluation.	Rate and Review by Instructional Nudges
Most Valuable Point (MVP)	In <i>Most Valuable Point (MVP)</i> , a checking for understanding strategy, students review their notes and identify what they consider the most valuable point (MVP). They then explain their MVP by writing it down within a specified time. Students share their MVPs with peers in small groups or with the entire class, promoting reflection, synthesis, and collaborative discussion.	-

Strategies to Check for Understanding	Brief Description	Resources
Muddiest Point	<i>Muddiest Point</i> is a strategy that allows students to reflect on their understanding and identify areas of confusion. By focusing on their muddiest point, students engage in metacognitive practice, while teachers gain insight into students' struggles, enabling early and targeted intervention.	Muddiest Point by McCormick Teaching Excellence Institute
Confidence Meter / Fist to Five / Traffic Light / Thumbs Up, Thumbs Down	Using a <i>confidence meter</i> , hand gestures, or traffic light signals allows students to communicate their level of understanding or confidence after completing a task. This strategy provides teachers with immediate feedback, enabling targeted support and intervention.	Confidence Meter by Instructional Nudges Book - Fast and Effective Assessment: How to Reduce Your Workload and Improve Student Learning, 2018, p.64-98
Around the Room	<i>Around the Room</i> is a review activity in which questions are placed around the classroom for students to select and attempt, either individually or in pairs or small groups.	Around the Room by Teaching with Jennifer Findley
Flash Cards	Using flashcards and quizzes helps students actively retrieve knowledge, reinforcing learning and strengthening memory retention.	Three Tips on How to Study Effectively by TED-Ed
Vocabulary Knowledge Rating (VKR)	In <i>Vocabulary Knowledge Rating (VKR)</i> , students assess their familiarity with specific terms before learning new content and continue to monitor their understanding throughout the learning process.	VKR by Dr. Andy Johnson
Up & Down Game	The <i>Up & Down Game</i> is an interactive quiz used to quickly assess students' understanding. Students indicate their responses by standing up or sitting down, providing immediate feedback to the teacher.	Book - Fast and Effective Assessment: How to Reduce Your Workload and Improve

Strategies to Check for Understanding	Brief Description	Resources
		Student Learning, 2018, p.64-98 28 Ways to Quickly Check for Understanding by Daniel Leonard, Edutopia
Grade As You Go	The teacher grades and assesses students' work while circulating around the classroom, allowing for real-time monitoring of understanding and providing immediate feedback.	5.3. Grade as You Go by Top Hat
Other Checking for Understanding Strategies	Refer to resources for more Checking for Understanding Strategies	28 Ways to Quickly Check for Understanding by Daniel Leonard, Edutopia 53 Ways to Check for Understanding by Edutopia

Appendix C

Specific Positive Classroom Culture Strategies Based on Primary Cause of Low Will (Jackson, 2011, pp.78)

Reasons for Resisting Learning	Specific Positive Classroom Culture Strategies to Try
Fear of Failure	● Link the currencies students have to the currencies the investment requires. ● Treat failure as an opportunity. ● Plan ahead to develop currencies before students need them. ● Build in more opportunities to practice. ● Provide detailed feedback designed to help students learn from mistakes ● Allow conditional retakes, assignment resubmissions, and opportunities for students to have some say over the difficulty level of their assignments.
Lack of Relevance	● Point out real relevance. ● Allow students to learn in ways that are most effective for them. ● Show students how the investment will matter to them. ● Stress long-term gains. ● Generate curiosity with story structure and by creating disequilibria. ● Break long-term tasks into smaller ones with payoffs in currencies that students value. ● Demystify soft skills. ● Focus students on the immediate, emotional rewards of investing.
Lack of Trust	● Invest in them first. ● Communicate clear goals with high expectations. ● Create support structures that help students invest successfully. ● Show students how to leverage currencies they have. ● Give forward-focused, growth-oriented feedback. ● Use proactive intervention. ● Praise effort, not ability. ● Collect and respond to feedback. ● Explain classroom policies.

Note: Currencies are knowledge, soft skills, social skills, and network affiliations.